Solution Architecture

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Solution Architecture Diagram:

Presentation / Client Layer

Web Browser Interface

home.html · input.html · output.html

Responsive HTML5 + CSS3 + Jinja2 templates served by Flask

Users: Loan Applicant · Credit Officer / Financial Analyst

API Gateway / Flask Application Layer

app.py — Flask Backend

Route: / · /predict · /submit

Routing · Form handling (POST) · Input validation · Prediction request processing · Result rendering

Input Validation	Core Logic Service	External / Deployment
Form Preprocessing	**ML Prediction Engine**	**IBM Cloud (Production)**
• Encode categorical fields • Cast to float • Build feature vector • Apply StandardScaler (scale1.pkl)	• XGBoost model loaded from rdf_model.json • Predicts: 0 (Declined) or 1 (Approved) • Trained on 614 rows • ~82% test accuracy	• Cloud hosting • Web access via browser • Local: Anaconda/Flask dev server (port 5000) • No external API dependencies

Data / Storage Layer

File-Based Model Artifacts · Training Dataset

loan_prediction.csv (614 rows × 13 cols — training data)

rdf_model.json (trained XGBoost model — native save format)

scale1.pkl (fitted StandardScaler for input normalization)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders home page, application input form, and prediction result page. Collects 11 applicant attributes via HTML form and sends them to the Flask backend via POST request.	HTML5, CSS3, JavaScript, Jinja2 templates
API Gateway / Flask Backend	Receives form POST requests at /submit, reads form values, builds a feature vector, scales it using the saved StandardScaler, passes it to the model, and renders the result back to the user.	Python 3.8+, Flask, Jinja2
Input Validation & Preprocessing	Converts form string values to float, builds a pandas DataFrame in the exact column order the model was trained on, and applies the saved StandardScaler transform before prediction.	pandas, NumPy, scikit-learn (StandardScaler)
ML Prediction Engine (Core Logic)	Loads the saved XGBoost model from rdf_model.json on startup. Accepts the scaled feature vector and returns a binary prediction: 1 = Approved, 0 = Declined. Trained on SMOTE-balanced data with ~82% test accuracy.	XGBoost (XGBClassifier), scikit-learn, imbalanced-learn (SMOTE)
Data / Storage Layer	Stores the historical training dataset (CSV), the trained model (JSON native format — avoids version mismatch errors caused by pickle), and the fitted scaler (pkl). No live database in this iteration.	loan_prediction.csv, rdf_model.json, scale1.pkl
Deployment Layer	Currently runs on a local Anaconda Flask development server (port 5000). Designed for production deployment on IBM Cloud per the original architecture specification.	Anaconda Navigator, Flask dev server (local), IBM Cloud (planned)